

MR Incorporation Website - Technical Documentation

Project Overview

MR Incorporation is a professional corporate website showcasing the company's journey from a humble garment trading business in 2003 to a distinguished holding company with operations across multiple countries. The website presents the company's evolution, organizational structure, global presence, and contact information through interactive visual elements.

Technologies & Tools Used

Frontend Technologies

- **HTML5:** Semantic markup and structure
- **CSS3:** Modern styling with animations and responsive design
- **Vanilla JavaScript (ES6+):** Interactive functionality and API integrations
- **SVG Graphics:** Scalable vector graphics for visual elements

External APIs & Services

- **Instagram Basic Display API:** For fetching and displaying Instagram posts
- **Gmail Web Interface:** For email functionality through captcha verification

Development Tools

- **Fetch API:** For loading external SVG files and Instagram data
 - **CSS Animations & Keyframes:** For smooth visual transitions
 - **CSS Grid & Flexbox:** For responsive layouts
 - **DOM Manipulation:** For dynamic content updates
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Key Features

1. Interactive Menu Wheel

- Animated navigation with zoom effects
- Smooth hover transitions
- Central navigation hub

2. Empowering Progress Animation

- Sequential burst animations for images and text
- Staggered timing effects
- Visual storytelling through animation

3. About Us Interactive Cards

- Hover-triggered information cards
- Dynamic positioning following mouse cursor
- Company history and milestones display

4. Organizational Structure Visualization

- Interactive organogram with detailed role descriptions
- Hover cards showing responsibilities
- Hierarchical structure display

5. Global Presence Map

- Interactive world map with hover effects
- Color-coded regions (current business vs expandable countries)
- Zoom and highlighting effects

6. Instagram Integration Gallery

- Real-time Instagram post fetching
- Auto-sliding carousel functionality
- Modal view for detailed post inspection
- Responsive image display

7. Advanced Captcha System

- Multi-step verification process
- Various captcha types (math, color, pattern, image, slider, sequence, text)
- Progress tracking
- Anti-bot protection for email contact

8. Accessibility Features

- Skip navigation links
- Keyboard shortcuts (Alt + 1-5 for sections)
- ARIA labels and roles
- Focus management
- Screen reader compatibility

📁 File Structure & Components

Core Files

├── index.html	# Main HTML structure
├── style/	
│ └── style.css	# Main stylesheet with animations
├── scripts/	
│ ├── empowering.js	# Empowering progress animations
│ ├── menu.js	# Menu wheel functionality
│ ├── aboutUs.js	# About section hover cards
│ ├── history.js	# History section loading
│ ├── organogram.js	# Organizational chart interactions
│ ├── contactUs.js	# Captcha system and email functionality
│ ├── coglagamards.js	# Additional graphics loading
│ ├── slider.js	# Instagram gallery functionality
│ └── map.js	# Interactive world map
└── svg/	# SVG graphics directory

🔍 Section Breakdown & Functionality

1. Header Section (Menu Wheel & Empowering)

Location: `.first-part` **Components:**

- Menu wheel with zoom hover effect
- Empowering progress animation

Hover Behavior:

- Menu wheel scales to 0.9x on hover with smooth transition
- Empowering elements appear with sequential burst animations

Key Features:

- Responsive 50-50 layout
- Smooth CSS transitions
- Sequential animation delays

2. About Us Section

Location: `#about-container` **Components:**

- Four interactive information cards

- SVG-based visual elements

Hover Behavior:

```
// Cards follow mouse cursor on hover
card.style.left = e.pageX + 15 + 'px';
card.style.top = e.pageY + 15 + 'px';
```

Content Areas:

- **Introduction Card:** Company founding and evolution
- **Orange Card:** Journey from university to Dubai headquarters
- **Green Card:** 21-year legacy and £200M investments
- **Yellow Card:** Future expansion plans (£450M commitment)

3. History Section

Location: #historyContainer **Components:**

- Historical timeline visualization
- Company milestones display

Functionality:

- Loads external SVG graphics
- Displays company evolution chronologically

4. Organizational Structure

Location: #organogramContainer **Components:**

- Interactive organizational chart
- 20+ role-specific hover cards

Hover Cards Include:

- Board of Directors (Central, Regional, Country)
- Executive positions (CEO, CFO)
- Committee structures (Executive, Working, Technical, Risk Management)
- Specialized roles (HR, Marketing, Tax, Equity)

Hover Behavior:

- Cards appear on role hover
- Detailed responsibility descriptions
- Dynamic positioning system

5. Global Presence Map

Location: #map-container **Components:**

- Interactive world map
- Two card types for different regions

Hover Effects:

```
#map-container path:hover {  
  fill: #4ad9eb;  
  transform: scale(1.05);  
  cursor: pointer;  
}
```

Card Types:

- **Current Business Card:** Shows established markets
- **Expandable Country Card:** Highlights growth opportunities

6. Instagram Gallery

Location: #gallerysection **Components:**

- Auto-sliding carousel
- Modal image viewer
- Instagram API integration

Functionality:

- Fetches 10 latest Instagram posts
- Auto-slides every 2 seconds
- Click-to-expand modal view
- Responsive image handling

API Integration:

```
const res = await  
fetch(`https://graph.instagram.com/me/media?fields=id,caption,media_type,media_url,thumbnail_url,timestamp,permalink&limit=10&access_token=${INSTAGRAM_ACCESS_TOKEN}`);
```

Features:

- Video thumbnail support
- Bengla date/time formatting
- Direct Instagram link access
- Loading error handling

7. Contact Section with Captcha System

Location: .contactUs **Components:**

- Multi-step captcha verification
- Email integration
- Progress tracking

Captcha Types:

1. **Mathematical:** Basic arithmetic operations
2. **Color Selection:** Visual color identification
3. **Pattern Recognition:** Shape selection
4. **Image/Emoji:** Emoji identification
5. **Slider:** Precise value setting
6. **Sequence:** Number pattern completion
7. **Text Input:** Word typing verification

Security Flow:

1. User clicks email link
2. Two-step captcha verification
3. Progress bar indicates completion
4. Successful verification opens Gmail compose

Validation Logic:

```
validateAnswer(captcha, answer) {  
  if (captcha.type === "slider") {  
    return Math.abs(parseInt(answer) - captcha.data.answer) <= 2;  
  }  
  return answer.toString().toLowerCase() ===  
  captcha.data.answer.toString().toLowerCase();  
}
```

Animation System

Empowering Progress Animations

Technique: CSS Keyframes with staggered delays **Duration:** 6.5 seconds infinite loop **Pattern:** Sequential burst effect for 23 text elements

```
@keyframes scaleInner {  
  0% { opacity: 0; transform: scale(0); }  
  80% { opacity: 1; transform: scale(0.9); }  
  100% { opacity: 1; transform: scale(1); }
```

}

Delay Pattern: 0.3s increments for text elements, 0.2s for images

Hover Animations

- **Menu Wheel:** Scale and transition effects
- **Map Regions:** Color change and zoom
- **Cards:** Smooth appearance with shadows
- **Buttons:** Transform and color transitions

Responsive Design

Breakpoints

- **Desktop:** 1300px max-width container
- **Tablet:** Flexible SVG scaling
- **Mobile:** 480px breakpoint for captcha system

Mobile Optimizations

- Reduced grid columns for captcha elements
- Flexible button layouts
- Optimized image sizes for gallery
- Touch-friendly interactive elements

& Accessibility Features

Keyboard Navigation

```
// Alt + Number shortcuts for section navigation
Alt + 1 → About Section
Alt + 2 → Organogram
Alt + 3 → Map
Alt + 4 → Gallery
Alt + 5 → Contact
```

Screen Reader Support

- ARIA labels for all major sections
- Skip navigation links

- Focus management in modals
- Semantic HTML structure

Visual Accessibility

- High contrast color schemes
 - Large clickable areas
 - Clear focus indicators
 - Alternative text for images
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Performance Optimizations

Loading Strategy

- **Lazy Loading:** SVG files loaded on demand
- **Progressive Enhancement:** Core functionality works without JavaScript
- **Image Optimization:** Instagram thumbnails for videos
- **Caching:** Browser caching for external resources

Animation Performance

- CSS transforms instead of layout changes
 - Hardware acceleration with `transform` properties
 - Optimized animation timing functions
 - Reduced DOM manipulation during animations
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Security Features

Captcha System Security

- **Multi-layered Verification:** Two-step process
- **Random Selection:** Prevents pattern recognition
- **Answer Validation:** Case-insensitive comparison
- **Session Management:** No persistent storage of answers
- **Anti-automation:** Various interaction types required

API Security

- **Instagram Token:** Secure token-based authentication
- **Error Handling:** Graceful failure handling

- **Rate Limiting:** Built-in API rate limiting
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Browser Compatibility

Supported Browsers

- **Chrome/Chromium:** Full support (90+)
- **Firefox:** Full support (85+)
- **Safari:** Full support (14+)
- **Edge:** Full support (90+)

Progressive Enhancement

- **Base Functionality:** Works without JavaScript
 - **Enhanced Experience:** Full interactivity with JavaScript
 - **Graceful Degradation:** Fallbacks for unsupported features
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Performance Metrics

Loading Performance

- **Initial Load:** ~2-3 seconds (depending on Instagram API)
- **SVG Loading:** Individual file loading on demand
- **Animation Performance:** 60fps smooth animations
- **Image Optimization:** Responsive image loading

User Experience

- **Hover Response:** Immediate (<100ms)
 - **Animation Smoothness:** Hardware-accelerated
 - **Mobile Touch:** Optimized touch targets
 - **Accessibility:** Full keyboard navigation support
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Future Enhancement Opportunities

Potential Improvements

1. **PWA Implementation:** Service worker for offline functionality
2. **Internationalization:** Multi-language support
3. **CMS Integration:** Dynamic content management
4. **Analytics Integration:** User behavior tracking
5. **SEO Optimization:** Meta tags and structured data
6. **Performance Monitoring:** Real-time performance tracking

Scalability Considerations

- Modular JavaScript architecture for easy maintenance
 - CSS custom properties for theme management
 - Component-based structure for reusability
 - API abstraction for easy service switching
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Maintenance Guidelines

Regular Updates Required

- **Instagram Access Token:** Refresh every 60 days
- **Browser Compatibility:** Test with latest browser versions
- **Security Patches:** Monitor for security vulnerabilities
- **Performance Monitoring:** Regular performance audits

Code Quality Standards

- **ES6+ JavaScript:** Modern JavaScript practices
- **CSS Best Practices:** BEM methodology consideration
- **Accessibility Compliance:** WCAG 2.1 guidelines
- **Cross-browser Testing:** Regular compatibility testing

This documentation provides a comprehensive overview of the MR Incorporation website's technical implementation, features, and maintenance requirements.